Experiment 2: Introduction to C Programming Environment

Objective: To familiarize with the basic C programming environment, including writing, compiling, and executing a simple program.

Questions:

1. **Write a C program to print "Hello, World!" on the screen.**
   Sample Input: `None`
   Sample Output:
   ```
   Hello, World!
   ```

2. **Write a program to declare integer, float, and character variables, assign them values, and print them.**
   Sample Input: `None (values are hard-coded)`
   Sample Output:
   ```
   Integer value: 10
   Float value: 25.500000
   Character value: A
   ```

3. **Write a program to find the area of a triangle.**
   Sample Input:
   ```
   Enter base (in meters): 7
   Enter height (in meters): 20
   ```
   Sample Output:
   ```
   Area of the triangle is 70 sq. meters
   ```

4. **Write a C program to calculate the area and perimeter of a circle. Take the radius as input.**
   Sample Input:
   ```
   Enter the radius of the circle: 7
   ```
   Sample Output:
   ```
   Perimeter of the circle is: 43.96
   Area of the circle is: 153.94
   ```

5. **Write a program to convert temperature from Fahrenheit to Celsius.**
   Sample Input:
   ```
   Enter temperature in Fahrenheit: 98.6
   ```
   Sample Output:
   ```
   Temperature in Celsius is: 37.00
   ```

6. **Write a program to calculate the amount payable if money has been lent on simple interest.**
   Sample Input:
   ```
   Enter Principal Amount: 25000
   Enter ROI: 5%
   Enter Time (in years): 5
   ```
   Sample Output:

```
Amount Payable (after 5 years) : 31250
```

7. **Write a Program to calculate and display the volume and total surface area of a CUBE.**

   Sample Input:
   ```
   Enter the side of the cube (in cms): 4
   ```
   Sample Output:
   ```
   Volume of the cube = 64.00 cubic cms
   Total surface area of the cube = 96.00 square cms
   ```

8. **Write a program to swap the values of two variables without using a third variable.**

   Sample Input: `None (values are hard-coded)`
   ```
   a: 10
   b: 20
   ```
   Sample Output:
   ```
   Before swapping: a = 10, b = 20
   After swapping: a = 20, b = 10
   ```

9. **Write a program to take two numbers as input and display their sum, difference, product, quotient and average.**

   Sample Input:
   ```
   Enter first number: 10
   Enter second number: 5
   ```
   Sample Output:
   ```
   Sum = 15
   Difference = 5
   Product = 50
   Average = 7.500000
   Quotient = 2.00
   ```

10. **Write a program to display the size of every data type using the "sizeof" operator.**

    Sample Input: `No input required`

    Sample Output:
    ```
    Size of char: 1 byte
    Size of short int: 2 bytes
    Size of int: 4 bytes
    Size of long int: 8 bytes
    Size of float: 4 bytes
    Size of double: 8 bytes
    Size of long double: 16 bytes
    ```